$$C_{\phi q}^{(11)[3]\,i1\_i2\_} \to$$

$$\frac{1}{16\pi^2}\Big(\frac{1}{9} g_2^4 \mathrm{LF}_{3,0}[m_2]\,\delta_{i1i2}+\frac{1}{6} g_2^4 \mathrm{LF}_{4,-1}[m_2]\,\delta_{i1i2}-\frac{8}{45} g_2^4 \mathrm{LF}_{5,-2}[m_2]\,\delta_{i1i2}+\frac{1}{18}\sum_p g_2^4 \mathrm{LF}_{3,0}[m_{\tilde{l}}^{p}]$$

$$\delta_{i1i2}-\frac{5}{48}\sum_p g_2^4 \mathrm{LF}_{4,-1}[m_{\tilde{l}}^{p}]\,\delta_{i1i2}+\frac{2}{45}\sum_p g_2^4 \mathrm{LF}_{5,-2}[m_{\tilde{l}}^{p}]\,\delta_{i1i2}+\frac{1}{6}\sum_p g_2^4 \mathrm{LF}_{3,0}[m_{\tilde{q}}^{p}]\,\delta_{i1i2}-$$

$$\frac{5}{16}\sum_p g_2^4 \mathrm{LF}_{4,-1}[m_{\tilde{q}}^{p}]\,\delta_{i1i2}+\frac{2}{15}\sum_p g_2^4 \mathrm{LF}_{5,-2}[m_{\tilde{q}}^{p}]\,\delta_{i1i2}+\frac{1}{18} g_2^4 \mathrm{LF}_{3,0}[\tilde{\mu}]\,\delta_{i1i2}+$$

$$\frac{1}{12} g_2^4 \mathrm{LF}_{4,-1}[\tilde{\mu}]\,\delta_{i1i2}-\frac{4}{45} g_2^4 \mathrm{LF}_{5,-2}[\tilde{\mu}]\,\delta_{i1i2}+\frac{1}{864} g_1^2 g_2^2 \mathrm{LF}_{2,1,0}[m_1, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}+$$

$$\frac{1}{864} g_1^2 g_2^2 \mathrm{LF}_{2,2,-1}[m_1, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}-\frac{1}{432} g_1^2 g_2^2 \mathrm{LF}_{3,1,-1}[m_1, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}+$$

$$\frac{1}{864} g_1^2 g_2^2 \mathrm{LF}_{4,1,-2}[m_1, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}+\frac{1}{864} g_1^2 g_2^2 \mathrm{LF}_{2,1,0}[m_1, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}+$$

$$\frac{1}{864} g_1^2 g_2^2 \mathrm{LF}_{2,2,-1}[m_1, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}-\frac{1}{432} g_1^2 g_2^2 \mathrm{LF}_{3,1,-1}[m_1, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}+$$

$$\frac{1}{864} g_1^2 g_2^2 \mathrm{LF}_{4,1,-2}[m_1, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}+\frac{2}{9} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,1,0}[m_1, m_{\tilde{u}}^{p}]-$$

$$\frac{4}{9} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{3,1,-1}[m_1, m_{\tilde{u}}^{p}]+\frac{2}{9} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{4,1,-2}[m_1, m_{\tilde{u}}^{p}]-$$

$$\frac{1}{96} g_2^4 \mathrm{LF}_{2,1,0}[m_2, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}-\frac{1}{96} g_2^4 \mathrm{LF}_{2,2,-1}[m_2, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}-\frac{5}{48} g_2^4 \mathrm{LF}_{3,1,-1}[m_2, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}+$$

$$\frac{1}{32} g_2^4 \mathrm{LF}_{4,1,-2}[m_2, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}-\frac{1}{96} g_2^4 \mathrm{LF}_{2,1,0}[m_2, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}-\frac{1}{96} g_2^4 \mathrm{LF}_{2,2,-1}[m_2, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}-$$

$$\frac{5}{48} g_2^4 \mathrm{LF}_{3,1,-1}[m_2, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}+\frac{1}{32} g_2^4 \mathrm{LF}_{4,1,-2}[m_2, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}-\frac{1}{3} g_2^4 \mathrm{LF}_{2,1,0}[m_2, \tilde{\mu}]\,\delta_{i1i2}-$$

$$\frac{1}{3} g_2^4 \mathrm{LF}_{2,2,-1}[m_2, \tilde{\mu}]\,\delta_{i1i2}+\frac{1}{6} g_2^4 \mathrm{LF}_{3,1,-1}[m_2, \tilde{\mu}]\,\delta_{i1i2}+\frac{1}{6} g_2^4 \mathrm{LF}_{3,2,-2}[m_2, \tilde{\mu}]\,\delta_{i1i2}+$$

$$\frac{1}{18} g_2^2 g_3^2 \mathrm{LF}_{2,1,0}[m_3, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}+\frac{1}{18} g_2^2 g_3^2 \mathrm{LF}_{2,2,-1}[m_3, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}-$$

$$\frac{1}{9} g_2^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_3, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}+\frac{1}{18} g_2^2 g_3^2 \mathrm{LF}_{4,1,-2}[m_3, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}+$$

$$\frac{1}{18} g_2^2 g_3^2 \mathrm{LF}_{2,1,0}[m_3, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}+\frac{1}{18} g_2^2 g_3^2 \mathrm{LF}_{2,2,-1}[m_3, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}-$$

$$\frac{1}{9} g_2^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_3, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}+\frac{1}{18} g_2^2 g_3^2 \mathrm{LF}_{4,1,-2}[m_3, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}+$$

$$\frac{2}{3} g_3^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,1,0}[m_3, m_{\tilde{u}}^{p}]-\frac{4}{3} g_3^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{3,1,-1}[m_3, m_{\tilde{u}}^{p}]+$$

$$\frac{2}{3} g_3^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{4,1,-2}[m_3, m_{\tilde{u}}^{p}]+\frac{1}{8} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}]\,\delta_{i1i2}-$$

$$\frac{1}{6} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pr} \mathrm{LF}_{5,1,-2}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}]\,\delta_{i1i2}+\frac{3}{8} g_2^2 \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}]\,\delta_{i1i2}-$$

$$\frac{1}{2} g_2^2 \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{pr} \mathrm{LF}_{5,1,-2}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}]\,\delta_{i1i2}+\frac{1}{4} g_2^2 \overline{a_u}^{pr} a_u^{pr} \mathrm{LF}_{2,2,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}]\,\delta_{i1i2}-$$

$$\frac{1}{8} g_2^2 \overline{a_u}^{pr} a_u^{pr} \mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}]\,\delta_{i1i2}-\frac{1}{432} g_1^2 g_2^2 \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1}, m_1]\,\delta_{i1i2}+$$

$$\frac{1}{864} g_1^2 g_2^2 \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1}, m_1]\,\delta_{i1i2}+\frac{1}{48} g_2^4 \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1}, m_2]\,\delta_{i1i2}-$$

$$\frac{1}{96} g_2^4 \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1}, m_2]\,\delta_{i1i2}-\frac{1}{9} g_2^2 g_3^2 \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1}, m_3]\,\delta_{i1i2}+$$

$$\frac{1}{18} g_2^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1}, m_3]\,\delta_{i1i2}-\frac{1}{432} g_1^2 g_2^2 \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2}, m_1]\,\delta_{i1i2}+$$

$$\frac{1}{864} g_1^2 g_2^2 \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2}, m_1]\,\delta_{i1i2}+\frac{1}{48} g_2^4 \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2}, m_2]\,\delta_{i1i2}-$$

$$\frac{1}{96} g_2^4 \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2}, m_2]\,\delta_{i1i2}-\frac{1}{9} g_2^2 g_3^2 \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2}, m_3]\,\delta_{i1i2}+$$

$$\frac{1}{18} g_2^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2}, m_3]\,\delta_{i1i2}-\frac{1}{4} g_2^2 \overline{a_u}^{pr} a_u^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2}-$$

$$\frac{1}{4} g_2^2 \overline{a_u}^{pr} a_u^{pr} \mathrm{LF}_{3,2,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2}+\frac{3}{4} g_2^2 \overline{a_u}^{pr} a_u^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2}-$$

$$\frac{1}{2} g_2^2 \overline{a_u}^{pr} a_u^{pr} \mathrm{LF}_{5,1,-2}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2}-\frac{5}{24} g_1^2 g_2^2 \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_1]\,\delta_{i1i2}+$$

$$\frac{1}{6} g_1^2 g_2^2 \mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_1]\,\delta_{i1i2}+\frac{2}{3} g_2^4 \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_2]\,\delta_{i1i2}+$$

$$\frac{1}{3} g_2^4 \mathrm{LF}_{3,2,-2}[\tilde{\mu}, m_2]\,\delta_{i1i2}-\frac{9}{8} g_2^4 \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_2]\,\delta_{i1i2}+\frac{1}{2} g_2^4 \mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_2]\,\delta_{i1i2}-$$

$$\frac{1}{8} g_2^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_{\tilde{d}}^{p}]+\frac{1}{24} g_2^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_{\tilde{d}}^{p}]+$$

$$\frac{1}{2} \overline{y_u}^{rp} \overline{y_u}^{i2s} y_u^{rs} y_u^{i1p} \mathrm{LF}_{2,1,0}[\tilde{\mu}, m_{\tilde{q}}^{r}]-\overline{y_u}^{rp} \overline{y_u}^{i2s} y_u^{rs} y_u^{i1p} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_{\tilde{q}}^{r}]+$$

$$\frac{1}{2} \overline{y_u}^{rp} \overline{y_u}^{i2s} y_u^{rs} y_u^{i1p} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_{\tilde{q}}^{r}]-\frac{1}{8} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_{\tilde{u}}^{p}]+$$

$$\frac{1}{24} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_{\tilde{u}}^{p}]-\frac{1}{36} m_1 g_1^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}]+$$

$$\frac{1}{72} m_1 g_1^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}]+\frac{1}{36} m_1 g_1^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}]-$$

$$\frac{1}{36} m_1 g_1^2 y_u^{i1p} \overline{a_u}^{i2p} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}]+\frac{1}{72} m_1 g_1^2 y_u^{i1p} \overline{a_u}^{i2p} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}]+$$

$$\frac{1}{36} m_1 g_1^2 y_u^{i1p} \overline{a_u}^{i2p} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}]-\frac{1}{72} m_1 g_1^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{u}}^{p}, m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{72} m_1 g_1^2 y_u^{i1p} \overline{a_u}^{i2p} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{u}}^{p}, m_{\tilde{q}}^{i2}]-\frac{1}{2} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,1,1,-1}[m_1, m_{\tilde{u}}^{p}, \tilde{\mu}]+$$

$$\frac{1}{3} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{3,1,1,-2}[m_1, m_{\tilde{u}}^{p}, \tilde{\mu}]+\frac{1}{16} g_1^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{d}}^{p}]-$$

$$\frac{1}{48} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i1}]-\frac{1}{48} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i2}]+$$

$$\frac{11}{48} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{u}}^{p}]+\frac{1}{16} g_2^4 \mathrm{LF}_{2,1,1,-1}[m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}]\,\delta_{i1i2}-$$

$$\frac{1}{8} g_2^4 m_2^2 \mathrm{LF}_{2,1,1,0}[m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}]\,\delta_{i1i2}+\frac{1}{16} g_2^4 \mathrm{LF}_{2,1,1,-1}[m_2, m_{\tilde{q}}^{i2}, \tilde{\mu}]\,\delta_{i1i2}-$$

$$\frac{1}{8} g_2^4 m_2^2 \mathrm{LF}_{2,1,1,0}[m_2, m_{\tilde{q}}^{i2}, \tilde{\mu}]\,\delta_{i1i2}-\frac{1}{16} g_2^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{d}}^{p}]+$$

$$\frac{1}{16}\left(3 g_2^2 \overline{y_u}^{i2p} y_u^{i1p}+g_2^4 \delta_{i1i2}\right) \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i1}]+\frac{1}{16} g_2^4 m_2^2$$

$$\mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}+\frac{1}{16}\left(3 g_2^2 \overline{y_u}^{i2p} y_u^{i1p}+g_2^4 \delta_{i1i2}\right) \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i2}]+$$

$$\frac{1}{16} g_2^4 m_2^2 \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}-\frac{1}{16} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{u}}^{p}]-$$

$$\frac{1}{3} m_3 g_3^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,1,1,0}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}]+\frac{1}{6} m_3 g_3^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}]+$$

$$\frac{1}{3} m_3 g_3^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{3,1,1,-1}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}]-\frac{1}{3} m_3 g_3^2 y_u^{i1p} \overline{a_u}^{i2p} \mathrm{LF}_{2,1,1,0}[m_3, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}]+$$

$$\frac{1}{6} m_3 g_3^2 y_u^{i1p} \overline{a_u}^{i2p} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}]+\frac{1}{3} m_3 g_3^2 y_u^{i1p} \overline{a_u}^{i2p} \mathrm{LF}_{3,1,1,-1}[m_3, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}]-$$

$$\frac{1}{6} m_3 g_3^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{u}}^{p}, m_{\tilde{q}}^{i1}]-\frac{1}{6} m_3 g_3^2 y_u^{i1p} \overline{a_u}^{i2p} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{u}}^{p}, m_{\tilde{q}}^{i2}]+$$

$$\frac{1}{16} \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{rp} \overline{y_u}^{rs} y_u^{i1s} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{p}, \tilde{\mu}, m_{\tilde{q}}^{r}]+\frac{1}{16} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{i1r} \overline{y_u}^{i2s} y_u^{ps}$$

$$\mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{r}, \tilde{\mu}, m_{\tilde{q}}^{p}]-\frac{1}{16} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{i1r} \overline{y_u}^{i2s} y_u^{ps} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, \tilde{\mu}, m_{\tilde{d}}^{r}]-$$

$$\frac{1}{16} \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{rp} \overline{y_u}^{rs} y_u^{i1s} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{r}, \tilde{\mu}, m_{\tilde{d}}^{p}]-\frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,0}[\tilde{\mu}, m_1, m_{\tilde{d}}^{p}]+$$

$$\frac{1}{16} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i1}]-\frac{1}{24} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i1}]+$$

$$\frac{1}{16} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i2}]-\frac{1}{24} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i2}]+$$

$$\frac{1}{16} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{u}}^{p}]+\frac{1}{8} g_2^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,0}[\tilde{\mu}, m_2, m_{\tilde{d}}^{p}]-$$

$$\frac{9}{16} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i1}]+\frac{3}{8} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i1}]-$$

$$\frac{9}{16} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i2}]+\frac{3}{8} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i2}]-$$

$$\frac{1}{16} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{u}}^{p}]-\frac{1}{8} \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{rp} \overline{y_u}^{rs} y_u^{i1s} \mathrm{LF}_{2,1,1,0}[\tilde{\mu}, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{r}]+$$

$$\frac{1}{8} \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{rp} \overline{y_u}^{rs} y_u^{i1s} \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{r}]-$$

$$\frac{1}{8} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{i1r} \overline{y_u}^{i2s} y_u^{ps} \mathrm{LF}_{2,1,1,0}[\tilde{\mu}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}]+$$

$$\frac{1}{8} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{i1r} \overline{y_u}^{i2s} y_u^{ps} \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}]-\frac{1}{48} g_1^2 g_2^2 \mathrm{LF}_{1,1,1,1,-1}[m_1, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}]\,\delta_{i1i2}-$$

$$\frac{1}{24} m_1 m_2 g_1^2 g_2^2 \mathrm{LF}_{1,1,1,1,0}[m_1, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}]\,\delta_{i1i2}-\frac{1}{48} g_1^2 g_2^2 \mathrm{LF}_{1,1,1,1,-1}[m_1, m_2, m_{\tilde{q}}^{i2}, \tilde{\mu}]$$

$$\delta_{i1i2}-\frac{1}{24} m_1 m_2 g_1^2 g_2^2 \mathrm{LF}_{1,1,1,1,0}[m_1, m_2, m_{\tilde{q}}^{i2}, \tilde{\mu}]\,\delta_{i1i2}-\frac{1}{48} g_1^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p}$$

$$\mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}]+\frac{1}{96} g_1^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}]-$$

$$\frac{1}{48} g_1^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}]+\frac{1}{96} g_1^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p}$$

$$\mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}]+\frac{1}{48} m_1 g_1^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}]+$$

$$\frac{1}{96} m_1 g_1^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}]+\frac{1}{48} m_1 g_1^2 y_u^{i1p} \overline{a_u}^{i2p}$$

$$\mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}]+\frac{1}{96} m_1 g_1^2 y_u^{i1p} \overline{a_u}^{i2p} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}]+$$

$$\frac{1}{16} g_2^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}]-\frac{1}{32} g_2^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p}$$

$$\mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}]+\frac{1}{16} g_2^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}]-$$

$$\frac{1}{32} g_2^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}]+\frac{1}{16} m_2 g_2^2 \overline{y_u}^{i2p} a_u^{i1p}$$

$$\mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}]+\frac{1}{32} m_2 g_2^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}]+$$

$$\frac{1}{16} m_2 g_2^2 y_u^{i1p} \overline{a_u}^{i2p} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}]+\frac{1}{32} m_2 g_2^2 y_u^{i1p} \overline{a_u}^{i2p}$$

$$\mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}]-\frac{1}{144} g_1^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}]-$$

$$\frac{1}{96} g_1^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}]-\frac{1}{96} g_1^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p}$$

$$\mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_1, m_{\tilde{q}}^{i2}, \tilde{\mu}]+\frac{1}{16} g_2^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}]+$$

$$\frac{1}{32} g_2^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}]+\frac{1}{32} g_2^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p}$$

$$\mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_2, m_{\tilde{q}}^{i2}, \tilde{\mu}]-\frac{1}{3} g_3^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}]+$$

$$\frac{1}{96} g_1^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{d}}^{p}, \tilde{\mu}]+\frac{1}{96} m_1 g_1^2 \overline{y_u}^{i2p} a_u^{i1p}$$

$$\mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{u}}^{p}, \tilde{\mu}]-\frac{1}{32} g_2^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{d}}^{p}, \tilde{\mu}]+$$

$$\frac{1}{32} m_2 g_2^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{u}}^{p}, \tilde{\mu}]+\frac{1}{96} g_1^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p}$$

$$\mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i2}, m_1, m_{\tilde{d}}^{p}, \tilde{\mu}]+\frac{1}{96} m_1 g_1^2 y_u^{i1p} \overline{a_u}^{i2p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i2}, m_1, m_{\tilde{u}}^{p}, \tilde{\mu}]-$$

$$\frac{1}{32} g_2^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i2}, m_2, m_{\tilde{d}}^{p}, \tilde{\mu}]+\frac{1}{32} m_2 g_2^2 y_u^{i1p} \overline{a_u}^{i2p}$$

$$\mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i2}, m_2, m_{\tilde{u}}^{p}, \tilde{\mu}]-\frac{1}{144} g_1^2 \overline{a_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}]-$$

$$\frac{1}{96} m_1 g_1^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}]-\frac{1}{96} m_1 g_1^2 y_u^{i1p} \overline{a_u}^{i2p}$$

$$\mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_1, m_{\tilde{q}}^{i2}, \tilde{\mu}]+\frac{1}{16} g_2^2 \overline{a_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}]-$$

$$\frac{1}{32} m_2 g_2^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}]-\frac{1}{32} m_2 g_2^2 y_u^{i1p} \overline{a_u}^{i2p}$$

$$\mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_2, m_{\tilde{q}}^{i2}, \tilde{\mu}]-\frac{1}{3} g_3^2 \overline{a_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}]-$$

$$\frac{1}{48} g_1^2 g_2^2 \mathrm{LF}_{2,1,1,1,-2}[\tilde{\mu}, m_1, m_2, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}+\frac{1}{48} m_1 m_2 g_1^2 g_2^2$$

$$\mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}-\frac{1}{48} g_1^2 g_2^2 \mathrm{LF}_{2,1,1,1,-2}[\tilde{\mu}, m_1, m_2, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}+$$

$$\frac{1}{48} m_1 m_2 g_1^2 g_2^2 \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}-\frac{1}{96} g_1^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p}$$

$$\mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}]-\frac{1}{96} g_1^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}]-$$

$$\frac{1}{96} m_1 g_1^2 \overline{y_u}^{i2p} a_u^{i1p} \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}]-\frac{1}{96} m_1 g_1^2 y_u^{i1p} \overline{a_u}^{i2p}$$

$$\mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}]+\frac{1}{32} g_2^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}]+$$

$$\frac{1}{32} g_2^2 \tilde{\mu}^2 \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}]-\frac{1}{32} m_2 g_2^2 \overline{y_u}^{i2p} a_u^{i1p}$$

$$\mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}]-\frac{1}{32} m_2 g_2^2 y_u^{i1p} \overline{a_u}^{i2p} \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}]\Big)$$